

Activation Functions in Neural Networks - Practical Guide

Aim

To study and implement various activation functions used in Neural Networks and visualize their outputs using Python.

Objectives

- To understand activation functions used in neural networks.
- To implement Sigmoid, ReLU, Tanh, Leaky ReLU and Softmax functions.
- To visualize activation functions using graphs.
- To understand the role of activation functions in deep learning.

Theory

Activation functions are mathematical functions used in Artificial Neural Networks to introduce non-linearity into the model. Without activation functions, neural networks behave like simple linear models and cannot solve complex problems. Different activation functions are used depending on the type of neural network and problem.

Activation Functions

1. Sigmoid Function

Formula: $\sigma(x) = 1 / (1 + e^{-x})$

Output Range: 0 to 1

Used in binary classification.

2. ReLU Function

Formula: $f(x) = \max(0, x)$

Removes negative values and improves training speed.

3. Tanh Function

Formula: $\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$

Output Range: -1 to 1.

4. Leaky ReLU Function

Formula: $f(x) = x$ if $x > 0$ else $0.01x$

Solves ReLU dead neuron problem.

5. Softmax Function

Formula: $\text{Softmax}(x_i) = e^{x_i} / \sum e^{x_j}$

Converts values into probabilities for multi-class classification.

Outcome

- Successfully implemented activation functions.
- Visualized graphs of activation functions.
- Understood practical applications of activation functions.

Conclusion

Activation functions are essential components of neural networks. They help neural networks learn complex patterns by introducing non-linearity. Different activation functions are suitable for different machine learning tasks.

How to Run .py File in Anaconda

- Step 1:** Open Anaconda Navigator.
- Step 2:** Open Spyder or VS Code from Anaconda.
- Step 3:** Create a new Python file.
- Step 4:** Paste the Python code.
- Step 5:** Save the file with .py extension.
- Step 6:** Click Run button or press F5.
- Step 7:** Graphs and outputs will appear.

How to Run .ipynb File in Anaconda

- Step 1:** Open Anaconda Navigator.
- Step 2:** Launch Jupyter Notebook.
- Step 3:** Open the .ipynb file.
- Step 4:** Click "Run All Cells".
- Step 5:** Observe outputs and graphs.

Code Explanation

- NumPy is used for mathematical calculations.
- Matplotlib is used for graph plotting.
- `np.linspace()` generates values from -10 to 10.
- Each activation function is calculated separately.
- `plt.plot()` draws graphs.
- `plt.show()` displays graphs.
- Softmax output is shown using a bar graph.

Important Viva Questions with Answers

Q1. What is an activation function?

An activation function introduces non-linearity into neural networks.

Q2. Why is ReLU widely used?

ReLU is computationally efficient and reduces vanishing gradient problems.

Q3. What is the output range of Sigmoid?

0 to 1.

Q4. Why is Softmax used?

Softmax converts outputs into probability distributions for multi-class classification.

Q5. What is the disadvantage of Sigmoid?

It suffers from vanishing gradient problems.

Q6. What is the output range of Tanh?

-1 to 1.

Q7. Which library is used for graph plotting?

Matplotlib.

Q8. Which activation function removes negative values?

ReLU.

Sample Code Snippet

```
# Activation Functions in Neural Networks

import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-10, 10, 200)

sigmoid = 1 / (1 + np.exp(-x))
relu = np.maximum(0, x)
tanh = np.tanh(x)
leaky_relu = np.where(x > 0, x, 0.01 * x)

plt.figure()
plt.plot(x, sigmoid)
plt.title("Sigmoid Function")
plt.grid()
plt.show()
```